



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Activity
24/07/26
Employee dataset
Name:E.Kavya
Reg NO:192424365

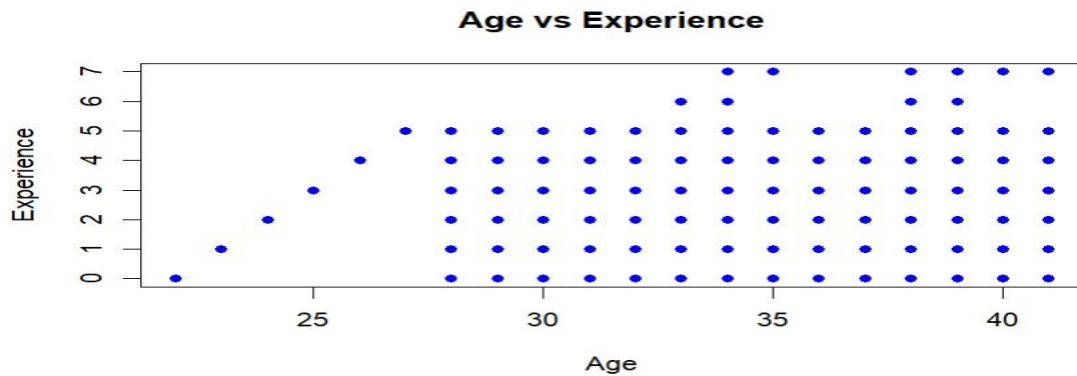
1. Scatter Plot:

A scatter plot is used to show the relationship between two numerical variables. Each point represents one observation, making it easy to identify patterns or correlations.

Code:

```
plot(employee$Age,  
      employee$ExperienceInCurrentDomain,  
      main="Age vs Experience",  
      xlab="Age",  
      ylab="Experience",  
      pch=19,  
      col="blue")
```

Output:



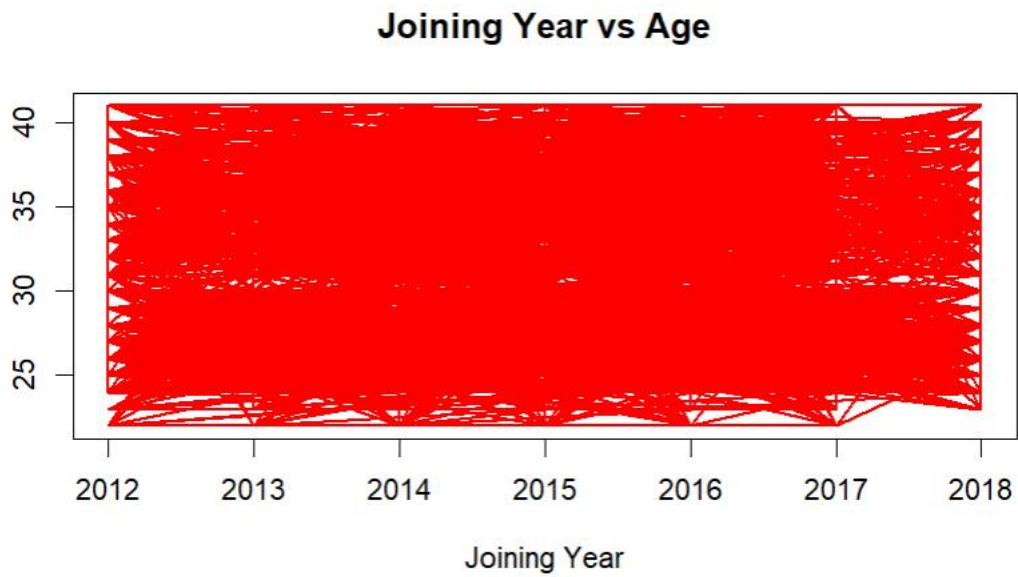
2. Line Chart

A line chart is used to display changes or trends in data over time or in a sequence. The data points are connected by lines to show continuous variation.

Code:

```
plot(employee$JoiningYear,  
      employee$Age,  
      type="l",  
      lwd=2,  
      col="red",  
      main="Joining Year vs Age",  
      xlab="Joining Year",  
      ylab="Age")
```

Output:



3. Bar Chart

A bar chart compares different categories using rectangular bars. The height of each bar represents the frequency or count of that category.

Code:

```
gender <- table(employee$Gender)

barplot(gender,

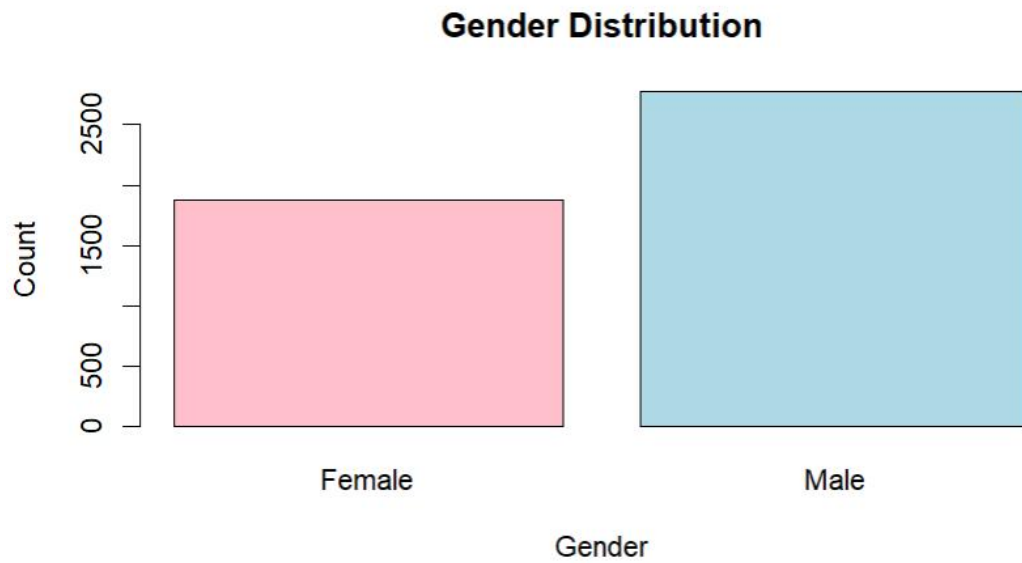
        main="Gender Distribution",

        xlab="Gender",

        ylab="Count",

        col=c("pink","lightblue"))
```

Output:



4. Pie Chart

A pie chart represents the proportion or percentage of each category as slices of a circle. It is useful for showing how the whole is divided into parts.

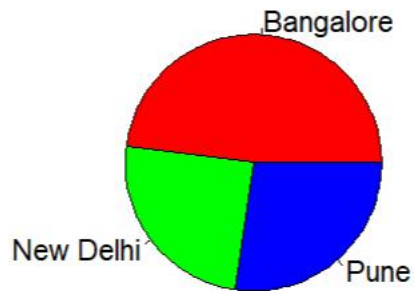
Code:

```
city <- table(employee$City)
```

```
pie(city,  
     main="Employees by City",  
     col=rainbow(length(city)))
```

Output:

Employees by City



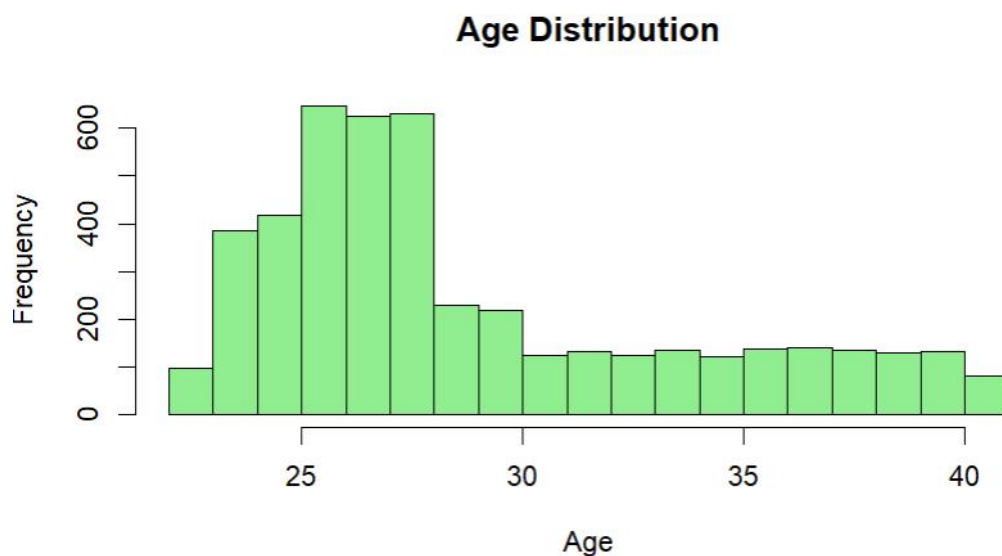
5. Histogram

A histogram displays the distribution of a numerical variable by grouping values into intervals called bins. It helps identify the shape and spread of the data.

Code:

```
hist(employee$Age,  
      main="Age Distribution",  
      xlab="Age",  
      col="lightgreen")
```

Output:



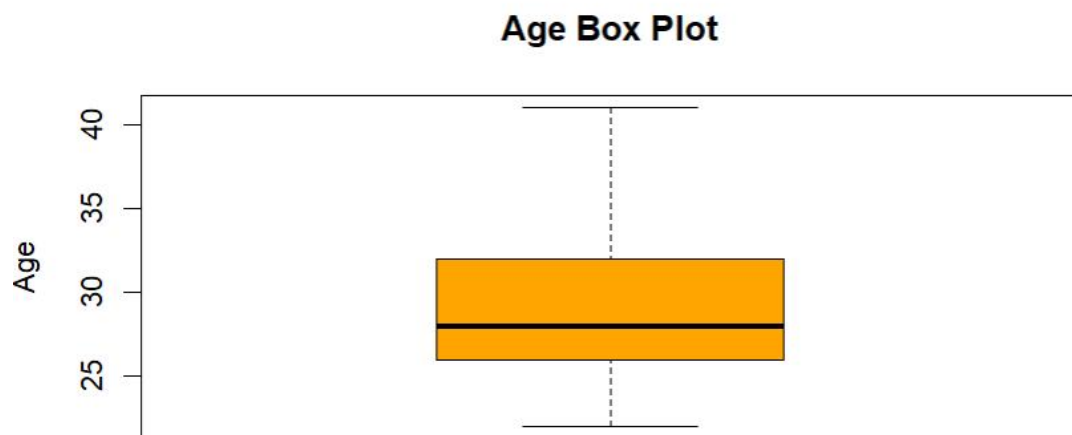
6. Box Plot

A box plot summarizes the distribution of data using the minimum, quartiles, median, and maximum values. It is useful for detecting outliers.

Code:

```
boxplot(employee$Age,  
        main="Age Box Plot",  
        ylab="Age",  
        col="orange")
```

Output:



7. Density Plot

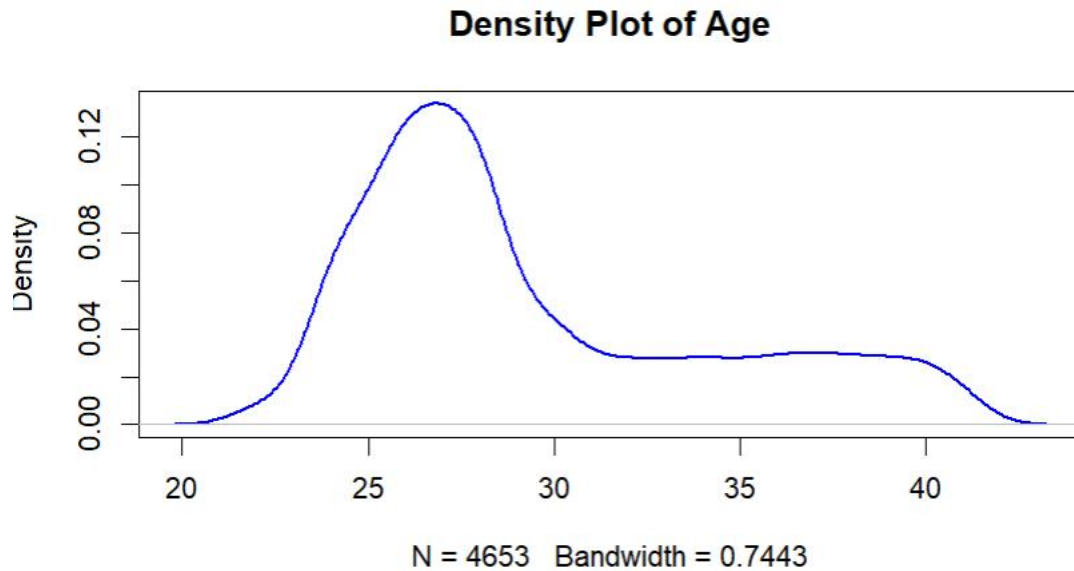
A density plot shows the distribution of a continuous variable using a smooth curve. It provides a clearer view of the data distribution than a histogram.

Code:

```
plot(density(employee$Age),  
     main="Density Plot of Age",  
     col="blue",
```

lwd=2)

Output:



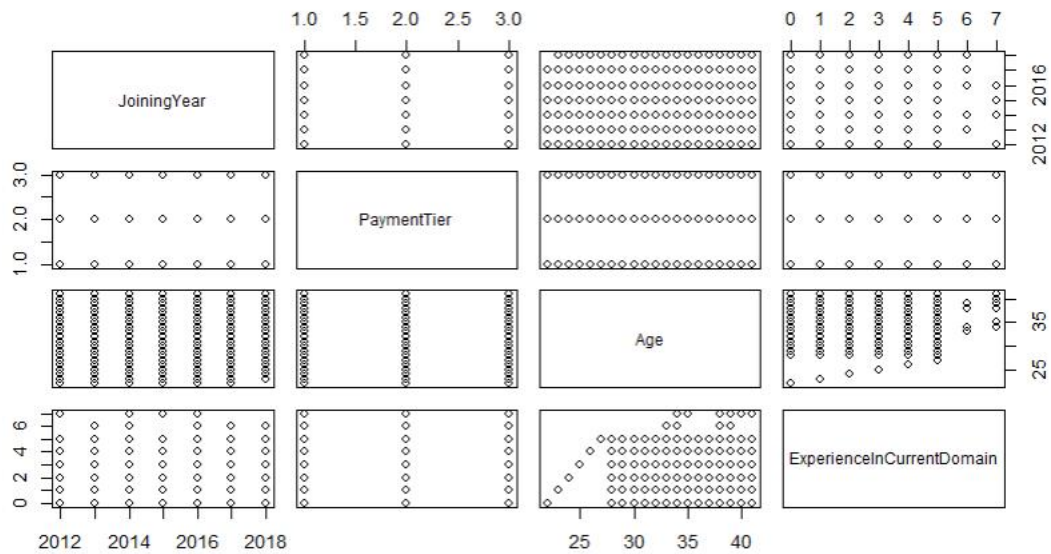
8. Scatter Plot Matrix (Pairs Plot)

A scatter plot matrix displays scatter plots for every pair of numerical variables. It is useful for identifying relationships among multiple variables simultaneously.

Code:

```
pairs(employee[,c("JoiningYear",  
                  "PaymentTier",  
                  "Age",  
                  "ExperienceInCurrentDomain")])
```

Output:



9.Stacked Bar Chart

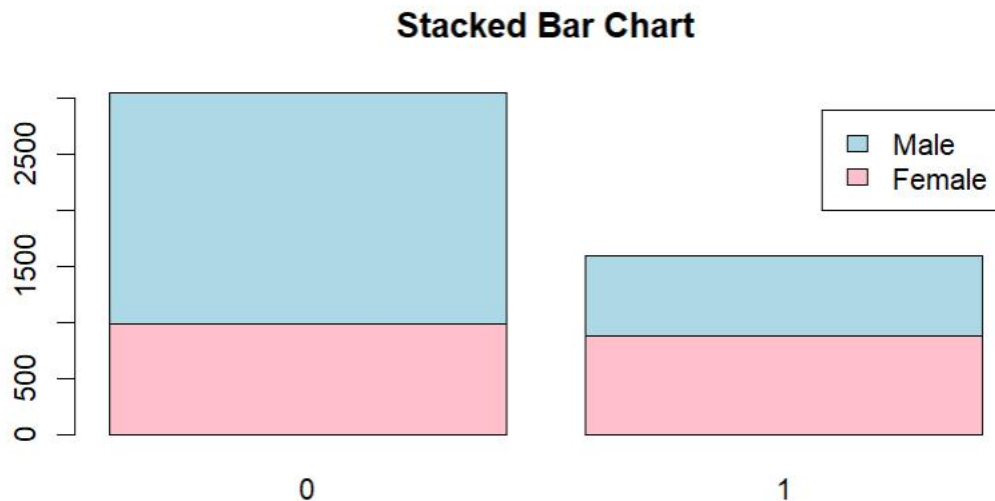
A stacked bar chart compares categories while also showing the contribution of subcategories within each bar.

Code:

```
data <- table(employee$Gender,
               employee$LeaveOrNot)

barplot(data,
        beside=FALSE,
        legend=TRUE,
        col=c("pink","lightblue"),
        main="Stacked Bar Chart")
```

Output:



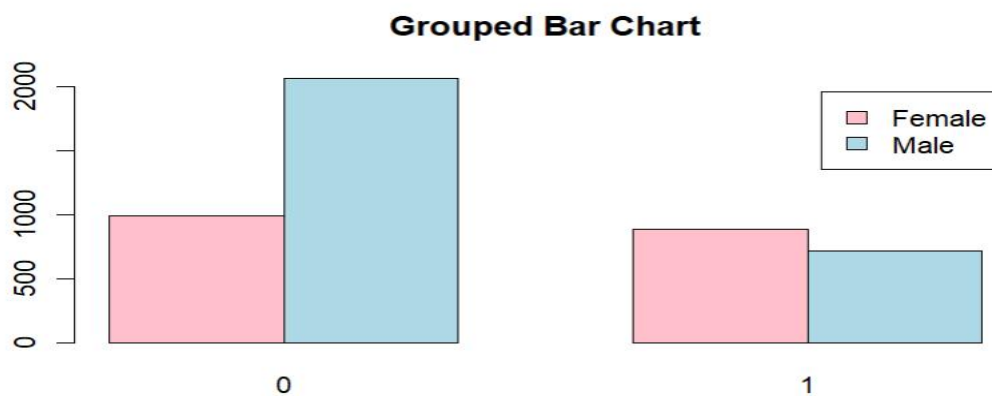
10. Grouped Bar Chart

A grouped bar chart places bars for different categories side by side. It makes it easy to compare multiple groups directly.

Code:

```
barplot(data,
        beside=TRUE,
        legend=TRUE,
        col=c("pink","lightblue"),
        main="Grouped Bar Chart")
```

Output:



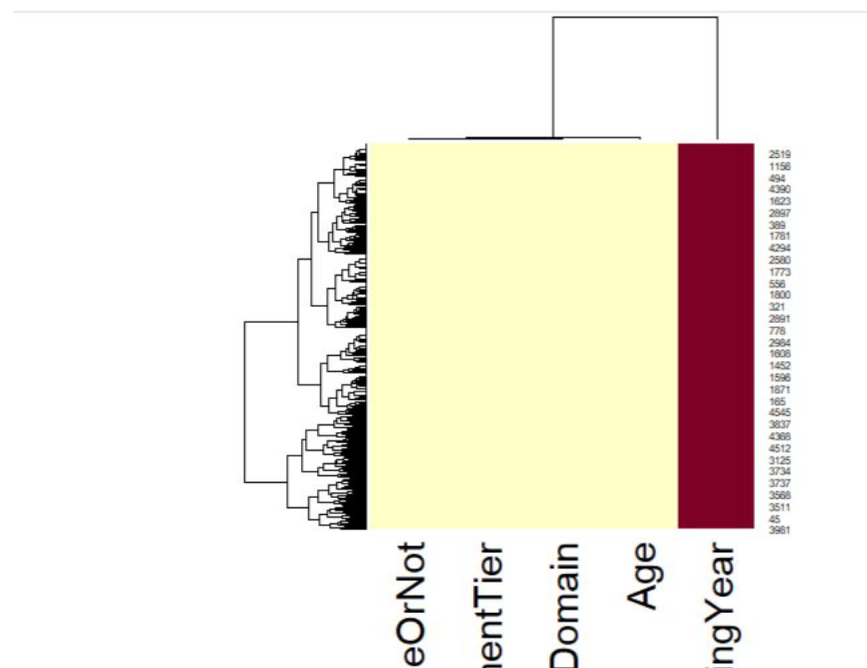
11. Heat Map

A heat map represents numerical values using different colors. Higher and lower values are distinguished by varying color intensities.

Code:

```
numdata <- employee[,c("JoiningYear",  
                        "PaymentTier",  
                        "Age",  
                        "ExperienceInCurrentDomain",  
                        "LeaveOrNot")]  
  
heatmap(as.matrix(numdata))
```

Output:



12. Correlation Matrix

A correlation matrix displays the correlation coefficients between numerical variables. It helps identify strong positive or negative relationships.

Code:

```
cor(numdata)
```

Output:

JoiningYear	1.00000000	-0.09607821	0.01316529	-0.03652462	0.01831432	-0.13464285
3652462						
PaymentTier	-0.09607821	1.00000000	0.00763090	0.01831432	-0.13464285	0.03050412
01831432						
Age	0.01316529	0.00763090	1.00000000	-0.03652462	0.01831432	-0.13464285
64285						
ExperienceInCurrentDomain	-0.03652462	0.01831432	-0.13464285	1.00000000	0.03050412	0.00000000
0.03050412						
LeaveOrNot	0.18170548	-0.19763777	-0.05112589	0.03050412	1.00000000	-0.00000000
03050412						

	LeaveOrNot
JoiningYear	0.18170548
PaymentTier	-0.19763777
Age	-0.05112589
ExperienceInCurrentDomain	-0.03050412
LeaveOrNot	1.00000000

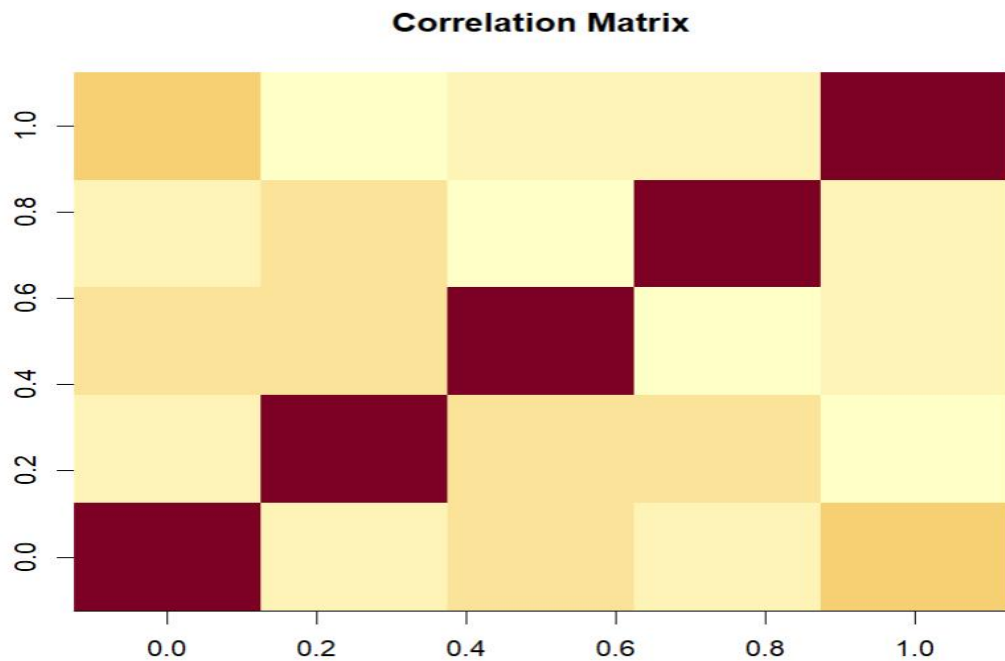
13. Correlation Heatmap

A correlation heatmap visualizes the correlation matrix using colors. Darker or brighter colors indicate stronger correlations.

Code:

```
image(cor(numdata),  
      main="Correlation Matrix")
```

Output:



14.. Area Chart

An area chart is similar to a line chart but fills the area beneath the line. It emphasizes the magnitude of values over time.

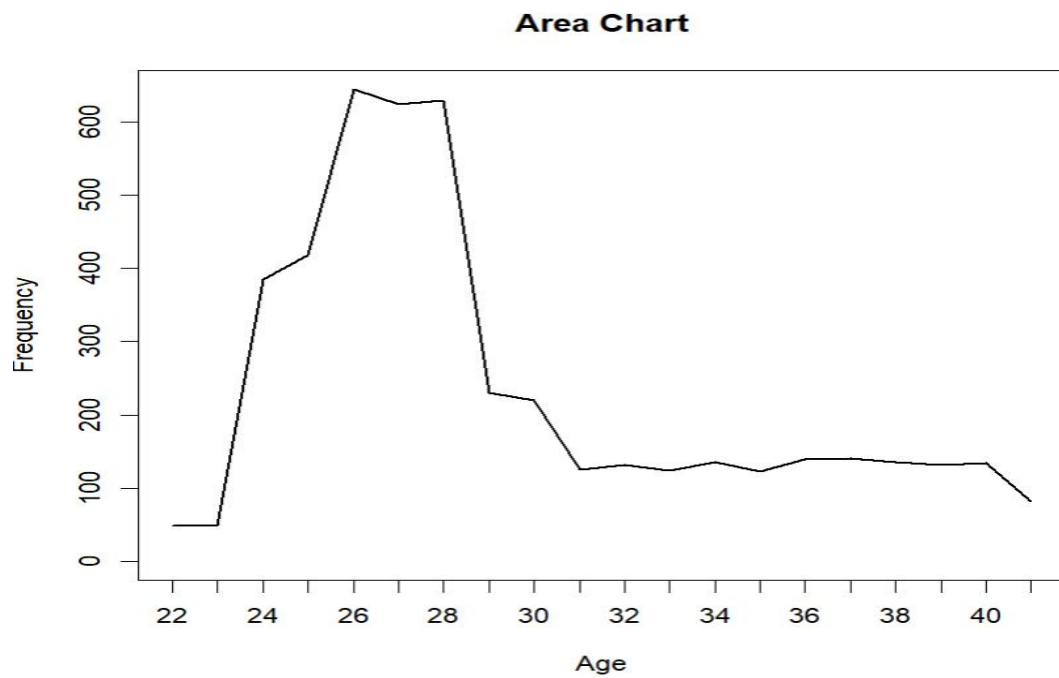
Code:

```
age <- table(employee$Age)

plot(age,
      type="l",
      main="Area Chart",
      xlab="Age",
      ylab="Frequency")

polygon(c(seq_along(age),length(age),1),
        c(age,0,0),
        col="lightblue")
```

Output:



15. Multiple Line Chart

A multiple line chart displays two or more lines on the same graph. It is useful for comparing trends of multiple variables over the same period.

Code:

```
plot(employee$JoiningYear,  
      employee$Age,  
      type="l",  
      col="red",  
      ylim=range(c(employee$Age,  
                    employee$ExperienceInCurrentDomain)),  
      xlab="Joining Year",  
      ylab="Value",  
      main="Multiple Line Chart")
```

```
lines(employee$JoiningYear,  
       employee$ExperienceInCurrentDomain,  
       col="blue")  
  
legend("topleft",  
       legend=c("Age","Experience"),  
       col=c("red","blue"),  
       lty=1)
```

Output:

